Problem D
Goldbach and Euler

Time limit: **1 second**
Memory limit: **1024 megabytes**

Problem Description

That every number which is resolvable into two prime numbers can be resolved into as many prime numbers as you like, can be illustrated and confirmed by an observation which you have formerly communicated to me, namely that every even number is a sum of two primes, and since $(n-2)$ is also a sum of two prime numbers, n must be a sum of three, and also four prime numbers, and so on. If, however, n is an odd number, then it is certainly a sum of three prime numbers, since $(n-1)$ is a sum of two prime numbers, and can therefore be resolved into as many prime numbers as you like. However, that every number is a sum of two primes, I consider a theorem which is quite true, although I cannot demonstrate it.

– ***Euler to Goldbach, 1742***

The above conjecture about all numbers being the sum of two primes (where 1 counts as a prime) is not always true, but it is more true for even numbers. Your task is to test the conjecture for specified integers, considering that prime numbers are the numbers which are positive and divisible by exactly two positive integers. Your program must be very efficient.

Input Format

The input file contains 100000 lines of input. Each line contains a single integer $n(0 < n \leq 100000000)$. Input is terminated by end of file.

Output Format

For each line of input produce one line of output. This line should be of one of the following types:

n `is not the sum of two primes!`
n `is the sum of` $p1$ `and` $p2$`.`

For the second case, always make sure that $(p2-p1)$ is positive and minimized.

Technical Specification

- $0 < n \leq 100000000$.

Sample Input 1

```
11
12
```

Sample Output 1

```
11 is not the sum of two primes!
12 is the sum of 5 and 7.
```

Sample Input 2

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
```

```
38
39
40
41
42
43
44
45
46
47
48
49
50
```

Sample Output 2

```
1 is not the sum of two primes!
2 is not the sum of two primes!
3 is not the sum of two primes!
4 is not the sum of two primes!
5 is the sum of 2 and 3.
6 is not the sum of two primes!
7 is the sum of 2 and 5.
8 is the sum of 3 and 5.
9 is the sum of 2 and 7.
10 is the sum of 3 and 7.
11 is not the sum of two primes!
12 is the sum of 5 and 7.
13 is the sum of 2 and 11.
14 is the sum of 3 and 11.
15 is the sum of 2 and 13.
16 is the sum of 5 and 11.
17 is not the sum of two primes!
18 is the sum of 7 and 11.
19 is the sum of 2 and 17.
20 is the sum of 7 and 13.
21 is the sum of 2 and 19.
22 is the sum of 5 and 17.
23 is not the sum of two primes!
24 is the sum of 11 and 13.
25 is the sum of 2 and 23.
26 is the sum of 7 and 19.
```

```
27 is not the sum of two primes!
28 is the sum of 11 and 17.
29 is not the sum of two primes!
30 is the sum of 13 and 17.
31 is the sum of 2 and 29.
32 is the sum of 13 and 19.
33 is the sum of 2 and 31.
34 is the sum of 11 and 23.
35 is not the sum of two primes!
36 is the sum of 17 and 19.
37 is not the sum of two primes!
38 is the sum of 7 and 31.
39 is the sum of 2 and 37.
40 is the sum of 17 and 23.
41 is not the sum of two primes!
42 is the sum of 19 and 23.
43 is the sum of 2 and 41.
44 is the sum of 13 and 31.
45 is the sum of 2 and 43.
46 is the sum of 17 and 29.
47 is not the sum of two primes!
48 is the sum of 19 and 29.
49 is the sum of 2 and 47.
50 is the sum of 19 and 31.
```

Sample Input 3

```
100000000
99999999
99999998
99999997
99999996
99999995
99999994
99999993
99999992
99999991
99999990
99999989
99999988
99999987
99999986
```

```
99999985
99999984
99999983
99999982
99999981
99999980
99999979
99999978
99999977
99999976
99999975
99999974
99999973
99999972
99999971
99999970
```

Sample Output 3

```
100000000 is the sum of 49999757 and 50000243.
99999999 is not the sum of two primes!
99999998 is the sum of 49999897 and 50000101.
99999997 is not the sum of two primes!
99999996 is the sum of 49999757 and 50000239.
99999995 is not the sum of two primes!
99999994 is the sum of 49999853 and 50000141.
99999993 is not the sum of two primes!
99999992 is the sum of 49999759 and 50000233.
99999991 is the sum of 2 and 99999989.
99999990 is the sum of 49999783 and 50000207.
99999989 is not the sum of two primes!
99999988 is the sum of 49999847 and 50000141.
99999987 is not the sum of two primes!
99999986 is the sum of 49999753 and 50000233.
99999985 is not the sum of two primes!
99999984 is the sum of 49999921 and 50000063.
99999983 is not the sum of two primes!
99999982 is the sum of 49999751 and 50000231.
99999981 is not the sum of two primes!
99999980 is the sum of 49999921 and 50000059.
99999979 is not the sum of two primes!
99999978 is the sum of 49999877 and 50000101.
```

```
99999977 is not the sum of two primes!
99999976 is the sum of 49999667 and 50000309.
99999975 is not the sum of two primes!
99999974 is the sum of 49999843 and 50000131.
99999973 is the sum of 2 and 99999971.
99999972 is the sum of 49999751 and 50000221.
99999971 is not the sum of two primes!
99999970 is the sum of 49999739 and 50000231.
```